

HODGE FILTRATIONS AND POLE-ORDER FILTRATIONS FOR SINGULAR HYPERSURFACES

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1. STATEMENT OF THE RESULTS

Let V be a hypersurface of degree d in projective space $\mathbf{P}^n$, and let μ be the integer part of n/d . If the base field is a finite field $\mathbf{F}_q$, a theorem of J. Ax [1], refining an earlier theorem of E. Warning, is equivalent to

$$(1.1) \quad \#V(\mathbf{F}_q) \equiv \sum_0^n q^i \pmod{q^\mu}.$$

It would amount to the same thing to take at the right-hand side the sum from 0 to $n-1$. If U is the complement $\mathbf{P}^n - V$ of V , this congruence can be rewritten more simply as

$$(1.2) \quad \#U(\mathbf{F}_q) \equiv 0 \pmod{q^\mu}.$$

In this article we prove a Hodge-theoretic analogue of (1.1) and (1.2). The analogy is based on Grothendieck's conjecture (extended to mixed motives) that if the de Rham realization of a motive M satisfies $M_{\text{DR}} = F^\mu M_{\text{DR}}$, then M is the tensor product of the Tate motive $\mathbf{Q}(-\mu)$ with an effective motive; consequently, after reduction modulo p to $\mathbf{F}_q$, the Frobenius eigenvalues should be divisible by q^μ . This conjecture, together with Dwork's work, lies behind Katz's conjecture (cf. [11, 2.9] and B. Mazur [13]).

Assume now that the base field is $\mathbf{C}$. In rational or complex cohomology, the restriction morphism

$$H^*(\mathbf{P}^n) \rightarrow H^*(V)$$

is injective, except in degree $i = 2n$. Indeed, its composite with the Gysin morphism $H^*(V) \rightarrow H^*(\mathbf{P}^n)$ is multiplication by the cohomology class of V , namely $d\eta$ where $\eta = c_1\mathcal{O}(1)$, and $H^*(\mathbf{P}^n) = \mathbf{Z}[\eta]/(\eta^{n+1})$.

The restriction morphism sits in a long exact sequence of mixed Hodge structures:

$$(1.3) \quad \cdots \rightarrow H_c^i(U) \rightarrow H^i(\mathbf{P}^n) \rightarrow H^i(V) \xrightarrow{\partial} H_c^{i+1}(U) \rightarrow \cdots$$

Our analogue of (1.1) is the following.

Theorem 1. *The cokernel $\text{coker}(H^*(\mathbf{P}^n) \rightarrow H^*(V))$ has Hodge filtration $\geq \mu$.*

A mixed Hodge structure H is said to have Hodge filtration $\geq \mu$ if $H = F^\mu H$. Exact sequence (1.3) shows that the theorem is equivalent to the following analogue of (1.2).

Variant 2 (1). *The compactly supported cohomology $H_c^*(U)$ has Hodge filtration $\geq \mu$.*

The variety U is smooth of pure dimension n . By Poincaré duality, $H_c^i(U)$ and $H^{2n-i}(U)$ are dual to one another, with values in the Hodge structure $\mathbf{Q}(-n)$ of pure Hodge type (n, n) . The theorem is therefore also equivalent to

Variant 3 (2). *The cohomology of U satisfies*

$$(1.4) \quad F^{n-\mu+1}H^*(U) = 0.$$

Date: Ann. Sci. É.N.S. (4) 23 (1990), 645–656
English translation and typeset working edition.

If V is the subvariety of $\mathbf{P}^n$ defined by equations $f_i = 0$ of degrees d_i ($1 \leq i \leq r$), J. Ax deduces from his theorem, applied to the hypersurface defined by the product of the f_i , that

$$(1.5) \quad \#V(\mathbf{F}_q) \equiv \sum_0^n q^i \pmod{q^\mu}$$

for

$$\mu = \left\lfloor \frac{n}{\sum d_i} \right\rfloor.$$

For the complement U of V , this is equivalent to

$$(1.6) \quad \#U(\mathbf{F}_q) \equiv 0 \pmod{q^\mu}.$$

Likewise, from Variant 1 of Theorem 1 and the spectral sequence defined by the open covering of U by the open sets $f_i \neq 0$ in $\mathbf{P}^n$, one deduces that over $\mathbf{C}$ the compactly supported cohomology of U has Hodge filtration $\geq [n/\sum d_i]$.

In the case of base field $\mathbf{F}_q$, N. Katz [11] proved that (1.5) and (1.6) remain true if $[n/\sum d_i]$ is replaced by the smallest integer $\geq (n+1-\sum d_i)/\sup(d_i)$. If d_1 is the largest of the d_i , this integer is

$$\left\lceil \frac{n - \sum_2^r d_i}{d_1} \right\rceil.$$

We do not know how to prove the Hodge-theoretic analogue of this refinement — except when V is smooth, in which case the result (SGA 7 XI 2.5) predates [11]. Added during revision: H. Esnault has just proved it when V has codimension r . Her approach is, roughly speaking, dual to ours, and has the advantage of requiring much less precise control over the desingularization procedure (cf. 2.8).

Let X be smooth over $\mathbf{C}$, let D be a divisor, let i denote the inclusion of D in X , and let $j : U \subset X$ be the inclusion of the open complement. By A. Grothendieck [8], one has

$$(1.7) \quad H^*(U, \Omega_U^*) \xrightarrow{\sim} H^*(U).$$

On the left, one uses the Zariski topology on U and the hypercohomology of the algebraic de Rham complex. On the right, cohomology means the complex cohomology of the usual topological space $U(\mathbf{C})$. The arrow is defined by the continuous map $\varepsilon : U(\mathbf{C}) \rightarrow U$ and the natural morphism from $\varepsilon^*\Omega_U^*$ to the resolution $\Omega_{U(\mathbf{C})}^*$ of the constant sheaf $\mathbf{C}$.

Since j is affine, one has $R^i j_* \Omega_U^p = 0$ for $i > 0$, hence $j_* \Omega_U^* \xrightarrow{\sim} Rj_* \Omega_U^*$ and therefore

$$(1.8) \quad H^*(X, j_* \Omega_U^*) \xrightarrow{\sim} H^*(U, \Omega_U^*).$$

The arrow is restriction to U . Altogether, one obtains

$$(1.9) \quad H^*(X, j_* \Omega_U^*) \xrightarrow{\sim} H^*(U).$$

For every subcomplex P of $j_* \Omega_U^*$ one thus has a map

$$H^*(X, P) \rightarrow H^*(U)$$

defined either via (1.9) or via (1.7) followed by restriction to U .

Each sheaf $j_* \Omega_U^p$ is the increasing union of the $\Omega_X^p(aD)$ ($a \geq 0$). The *pole-order filtration* P (or P_D , if the choice of D is ambiguous) on $j_* \Omega_U^*$ is the decreasing filtration defined by

$$P^s j_* \Omega_U^p = \begin{cases} 0, & p < s, \\ \Omega_X^p((p-s+1)D), & p \geq s. \end{cases}$$

It is decreasing, $P^s = 0$ for $s > \dim(X)$, and $j_* \Omega_U^*$ is the union of the P^s .

It is known that if D is smooth, the inclusion of filtered complexes

$$(1.10) \quad (\Omega_X^*(\log D), F) \hookrightarrow (j_* \Omega_U^*, P)$$

is a filtered quasi-isomorphism ([4, II, 3.13], inspired by [9]; the reader will also find there a more complicated analogue when D is a normal-crossings divisor).

Recall that for any normal-crossings divisor D , $\Omega_X^*(\log D) \subset j_*\Omega_X^*$ is the subsheaf of differential forms α such that both α and $d\alpha$ have at most simple poles along D . Another description, when D is smooth, is

$$\Omega_X^*(\log D) = \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_D^*)(D).$$

The Hodge filtration F is the stupid filtration by the truncations $\sigma_{\geq p}$.

Theorem 4. *Let X be smooth proper and let $D \subset X$ be a divisor. With the notation above, for every s , the image of $H^*(X, P^s(j_*\Omega_U^*))$ in $H^*(U, \mathbf{C})$ contains $F^s H^*(U, \mathbf{C})$.*

If D is smooth, it follows from the definitions and from the filtered quasi-isomorphism (1.10) that

$$H^*(X, P^s j_*\Omega_U^*) \xrightarrow{\sim} F^s H^*(U, \mathbf{C}).$$

Let us deduce Variant 2 of Theorem 1 from Theorem 2. It suffices to show that, with the notation of Theorem 1,

$$R\Gamma(\mathbf{P}^n, P^{n-\mu+1} j_*\Omega_U^*) = 0.$$

A fortiori, it suffices to show that for each p one has

$$R\Gamma(\mathbf{P}^n, P^{n-\mu+1} j_*\Omega_U^p) = 0.$$

The sheaves whose cohomology must be taken are either zero or of the form $\Omega^p(a)$ with $0 \leq p \leq n$, $a > 0$, and $a = (p - (n - \mu + 1) + 1)d$. It is known that $R\Gamma \Omega^p(a) = 0$ if $0 < a \leq p$ (R. Bott [3, Prop. 14.4, p. 246]; for an algebraic proof, valid in all characteristics, see SGA 7 XI 1.1). It remains to verify that

$$(1.11) \quad (p - (n - \mu + 1) + 1)d \leq p.$$

As p increases, the left-hand side grows faster than the right-hand side, so it is enough to consider the case $p = n$. Then (1.11) reduces to the inequality $\mu d \leq n$, which is precisely the definition of μ .

2. PROOF OF THEOREM 2

Proposition 5 (2.1). *Let X be smooth, $Y \subset X$ a smooth closed subscheme, i the inclusion of Y into X , $f : \tilde{X} \rightarrow X$ the blow-up of X along Y , and $\tilde{Y} = f^{-1}(Y) \subset \tilde{X}$ the exceptional divisor. Then on $\tilde{X}$ one has*

$$f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = \text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p)$$

and

$$R^i f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = 0 \quad \text{for } i > 0.$$

Proof. Using again i for the inclusion of $\tilde{Y}$ into $\tilde{X}$, one has

$$\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) = \text{Ker}(\Omega_{\tilde{X}}^p \rightarrow i_*\Omega_{\tilde{Y}}^p),$$

hence a morphism

$$\text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p) \rightarrow f_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}),$$

and it is enough to show that this induces an isomorphism from $\text{Ker}(\Omega_X^p \rightarrow i_*\Omega_Y^p)$, placed in degree 0, to $Rf_*\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y})$.

The proposition is local for the étale topology. It is therefore enough to treat the case where X is affine space $\mathbf{A}^{n_1+n_2} = \mathbf{A}^{n_1} \times \mathbf{A}^{n_2}$ and $Y = \{0\} \times \mathbf{A}^{n_2}$. Künneth formulas then reduce matters to the case $n_2 = 0$: the pair (X, Y) is obtained from $(\mathbf{A}^{n_1}, 0)$ by product with $\mathbf{A}^{n_2}$, and the direct sum over p of the sheaves under consideration is obtained from the corresponding sum for $(\mathbf{A}^{n_1}, 0)$ by external tensor product with $\bigoplus \Omega_{\mathbf{A}^{n_2}}^p$. We may therefore assume, and do assume, that Y is a point.

When Y is a point, we give two proofs of 2.1. The first has the advantage of working in every characteristic. The second explains why 2.1 is “obvious” from the point of view of Hodge theory.

First proof. Assume Y is a point. For $p > 0$, a local calculation shows that the morphism $f^*\Omega_X^p \rightarrow \Omega_{\tilde{X}}^p$ induces an isomorphism

$$f^*\Omega_X^p \xrightarrow{\sim} \text{Ker}(\Omega_{\tilde{X}}^p \rightarrow i_*\Omega_Y^p).$$

Since Ω_X^p is locally free and $\Omega_Y^p = 0$, the desired statement reduces to

$$\mathcal{O}_X \xrightarrow{\sim} Rf_*\mathcal{O}_{\tilde{X}}.$$

For $p = 0$, one uses the long exact sequence deduced from

$$0 \rightarrow \text{Ker}(\mathcal{O}_{\tilde{X}} \rightarrow i_*\mathcal{O}_{\tilde{Y}}) \rightarrow \mathcal{O}_{\tilde{X}} \rightarrow i_*\mathcal{O}_{\tilde{Y}} \rightarrow 0,$$

together with the same formula $\mathcal{O}_X \xrightarrow{\sim} Rf_*\mathcal{O}_{\tilde{X}}$ and the fact that for the projective space $\tilde{Y}$, the cohomology $H^*(\tilde{Y}, \mathcal{O}_{\tilde{Y}})$ is reduced to $\mathbf{C}$ in degree 0.

Second proof. The relation with Hodge theory is given by the following lemma.

Lemma 6 (2.2). *Let S be smooth projective, let $i : T \subset S$ be a smooth closed subscheme, and put $U := S - T$. The subcomplex $\text{Ker}(\Omega_S^* \rightarrow i_*\Omega_T^*)$ of the de Rham complex has hypercohomology equal to the compactly supported cohomology $H_c^*(U)$; the spectral sequence defined by the stupid filtration $\sigma_{\geq p}$ degenerates at E_1 and abuts to the Hodge filtration on $H_c^*(U)$.*

Apply this lemma to (X, Y) and $(\tilde{X}, \tilde{Y})$, with X smooth projective and Y a point. Because $\tilde{X} - \tilde{Y} \xrightarrow{\sim} X - Y$, one has $H_c^*(X - Y) \xrightarrow{\sim} H_c^*(\tilde{X} - \tilde{Y})$ and, passing to Gr_F , one obtains

$$(2.2.1) \quad R\Gamma(X, \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_Y^*)) \xrightarrow{\sim} R\Gamma(\tilde{X}, \text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*)).$$

The right-hand side is also

$$R\Gamma(X, Rf_*\text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*)).$$

Let C be the cone of the morphism

$$(2.2.2) \quad \text{Ker}(\Omega_X^* \rightarrow i_*\Omega_Y^*) \rightarrow Rf_*\text{Ker}(\Omega_{\tilde{X}}^* \rightarrow i_*\Omega_{\tilde{Y}}^*).$$

Applying $R\Gamma$ to the distinguished triangle defining it, one deduces from (2.2.1) that $R\Gamma C = 0$. On the other hand, C — or rather its cohomology sheaves — is supported at the point Y . From $R\Gamma C = 0$ one concludes that $C = 0$: morphism (2.2.2) is an isomorphism. This completes the second proof of 2.1, modulo 2.2.

Lemma 2.2 comes from the fact that the simple complex associated with the double complex $\Omega_S^* \rightarrow i_*\Omega_T^*$, together with the weight filtration

$$0 \subset W_{-1} = s(0 \rightarrow i_*\Omega_T^*) \subset W_0 = s(\Omega_S^* \rightarrow i_*\Omega_T^*),$$

and the Hodge filtration by the subcomplexes

$$s(\sigma_{\geq p}\Omega_S^* \rightarrow \sigma_{\geq p}i_*\Omega_T^*)$$

is a cohomological mixed Hodge complex. This implicitly includes the rational structure given by the filtered complex $\mathbf{Q} \rightarrow i_*\mathbf{Q}$ with $W^{-1} : 0 \rightarrow i_*\mathbf{Q}$.

The morphism

$$(\text{Ker}(\Omega_S^* \rightarrow i_*\Omega_T^*), \sigma_{\geq p}) \rightarrow (s(\Omega_S^* \rightarrow i_*\Omega_T^*), F)$$

is a filtered quasi-isomorphism. This simply expresses the fact that $\Omega_S^* \rightarrow i_*\Omega_T^*$ is an epimorphism. This suffices to establish 2.2 for a functorial mixed Hodge structure on $H_c^*(U)$. This statement, slightly less precise than 2.2, is enough for the application to 2.1. $\square$

2.3. Let X be smooth, let $V \subset X$ be a divisor, and let $Y \subset X$ be a smooth closed subscheme contained in the singular locus of V . Let $f : \tilde{X} \rightarrow X$ be the blow-up of X along Y , let $\tilde{Y} = f^{-1}(Y)$ be the exceptional divisor, and let $\tilde{V} = f^{-1}(V)$. Because Y is contained in the singular locus of V , one has

$$(2.3.1) \quad 2\tilde{Y} \leq \tilde{V}.$$

Let $V_1 = \tilde{V} - \tilde{Y}$. By (2.3.1) one has

$$(V_1)_{\text{red}} = (f^{-1}V)_{\text{red}}.$$

Let $U = X - V = \tilde{X} - \tilde{V}$, let j denote its inclusion into X , and let $\tilde{j}$ denote its inclusion into $\tilde{X}$. We write P for both pole-order filtrations, that on $j_*\Omega_U^*$ relative to V and that on $\tilde{j}_*\Omega_U^*$ relative to V_1 .

Corollary 7 (2.4). *With the hypotheses and notation of 2.3, there exists in the derived category a morphism*

$$(2.4.1) \quad Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow P^sj_*\Omega_U^*$$

inducing the identity on U .

Proof. The complex $P^s\tilde{j}_*\Omega_U^*$ has components

$$\Omega_{\tilde{X}}^p((p-s+1)V_1) \quad (p \geq s),$$

and, by (2.3.1), these are subsheaves of

$$\Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y})((p-s+1)\tilde{V}) = \Omega_{\tilde{X}}^p(\log \tilde{Y})(-\tilde{Y}) \otimes f^*\mathcal{O}((p-s+1)V).$$

By 2.1 the latter sheaves are acyclic: $R^if_* = 0$ for $i > 0$. Hence the derived direct image of the complex formed by them is obtained simply by applying f_* componentwise; this yields the complex of sheaves $\text{Ker}(\Omega_X^p \rightarrow \Omega_Y^p)((p-s+1)V)$.

Applying Rf_* to the inclusion

$$P^s\tilde{j}_*\Omega_U^* \rightarrow \sigma_{\geq s}(\Omega_{\tilde{X}}^*(\log \tilde{Y})(-\tilde{Y})((s-s+1)\tilde{V})),$$

one obtains a morphism

$$Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow \sigma_{\geq s} \text{Ker}(\Omega_X^* \rightarrow \Omega_Y^p)((s-s+1)V).$$

Composing with the inclusion of $\text{Ker}(\Omega_X^* \rightarrow \Omega_Y^*)$ into Ω_X^* gives the promised morphism

$$Rf_*P^s\tilde{j}_*\Omega_U^* \rightarrow P^sj_*\Omega_U^*.$$

□

Corollary 8 (2.5). *With the hypotheses and notation of 2.3, the image of $R^*\Gamma(\tilde{X}, P^s\tilde{j}_*\Omega_U^*)$ in $H^*(U)$ contains the image of $R^*\Gamma(X, P^sj_*\Omega_U^*)$.*

Proof. For any complex of sheaves K on $\tilde{X}$, one has $R\Gamma(X, Rf_*K) = R\Gamma(\tilde{X}, K)$. Applying $R\Gamma(X, \cdot)$ to 2.4 therefore yields a commutative diagram

$$\begin{array}{ccc} R^*\Gamma(\tilde{X}, P^s\tilde{j}_*\Omega_U^*) & \longrightarrow & R^*\Gamma(X, P^sj_*\Omega_U^*) \\ & \searrow & \swarrow \\ & H^*(U) = R^*\Gamma(U, \Omega_U^*) & \end{array}$$

□

2.6. By Hironaka [10], starting from (X, V) one can construct a sequence of blow-ups (X_i, V_i) ($0 \leq i \leq r$), with $(X_0, V_0) = (X, V)$, such that (X_{i+1}, V_{i+1}) is obtained from (X_i, V_i) in the same way as $(\tilde{X}, V_1)$ is obtained from (X, V) in 2.3, and in such a way that $(V_r)_{\text{red}}$ is a normal-crossings divisor in X_r . Let j_i be the inclusion of $U = X - V$ into X_i , and let P denote the pole-order filtration on $j_{i*}\Omega_U^*$ relative to the divisor V_i . Iterating 2.5, one finds that the image of $R^*\Gamma(X, P^s j_*\Omega_U^*)$ in $H^*(U)$ contains the image of $R^*\Gamma(X_r, P^s j_{r*}\Omega_U^*)$. In X_r , the open set U is the complement of the normal-crossings divisor $(V_r)_{\text{red}}$. By definition, $F^s H^*(U)$ is therefore the hypercohomology on X_r of the complex $\sigma_{\geq s}\Omega_{X_r}^*(\log(V_r)_{\text{red}})$.

One has

$$\sigma_{\geq s}\Omega_{X_r}^*(\log(V_r)_{\text{red}}) \subset P^s j_{r*}\Omega_U^*,$$

so $F^s H^*(U)$ is contained in the image of the hypercohomology of $P^s j_{r*}\Omega_U^*$ on X_r , and Theorem 2 follows.

Remark 9 (2.7). Just as 2.5 was deduced from 2.4 on X , one can refine Theorem 2 by a result on X : for X_r, V_r as in 2.6, there exists a morphism

$$Rf_* F^s \Omega_{X_r}^*(\log(V_r)_{\text{red}}) \rightarrow P^s j_* \Omega_U^*$$

that is the identity on U . More precisely, in the filtered derived category of X there is a morphism

$$Rf_*(\Omega_{X_r}^*(\log(V_r)_{\text{red}}), F) \rightarrow (j_* \Omega_U^*, P)$$

inducing the identity on U . The proof is the same, using 2.4 instead of 2.5.

A variant of results of Ph. Du Bois [7] also says that $Rf_*(\Omega_{X_r}^*(\log(V_r)_{\text{red}}), F)$ is independent of the choice of (X_r, V_r) dominating (X, V) , with X_r proper over X , $X_r - V_r \xrightarrow{\sim} X - V$, and such that $V_r = f^{-1}(Y)$ is a normal-crossings divisor in X_r .

Remark 10 (2.8). Our proof used the fact that in Hironaka's embedded resolution theorem one may choose the successive blow-ups with centers contained in the singular locus of $f^{-1}(V)$. This fact is not stated explicitly in [10], but it follows from the proof.

3. APPLICATIONS AND VARIANTS

3.1. Let X be smooth proper of pure dimension n , let D be a divisor, put $\mathcal{L} := \mathcal{O}(D)$, and set $U := X - D$. The arguments in Section 1 also show that if $H^*(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)}) = 0$ for $0 \leq i < \mu$, then $F^{n-\mu+1} H^*(U) = 0$. By Serre duality, the hypothesis may be reformulated as $H^*(X, \Omega^i \otimes \mathcal{L}^{\otimes-(\mu-i)}) = 0$ for $0 \leq i < \mu$. By Poincaré duality, the conclusion is equivalent to saying that $H_c^*(U)$ has Hodge filtration $\geq \mu$.

If one is interested in a single $H^m(U)$, it suffices to consider the groups $H^q(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)})$ with $q + n - i = m$. In particular, if U is affine, so that $H^m(U) = 0$ for $m > n$, it suffices to consider the groups $H^q(X, \Omega^{n-i} \otimes \mathcal{L}^{\otimes(\mu-i)})$ with $q + n - i \leq n$, i.e. $0 \leq q \leq i < \mu$. For $\mu = 1$ this gives:

Proposition 11 (3.2). *With the above notation, if U is affine and if $\Omega^n(D)$ has no nonzero global section, then $F^n H^*(U) = 0$.*

Here is a more direct proof of 3.2. Consider again $f : \tilde{X} \rightarrow X$ proper with $f^{-1}(U) \xrightarrow{\sim} U$, where $\tilde{X}$ is smooth and $\tilde{D} := f^{-1}(D)$ is a normal-crossings divisor. By definition,

$$F^n H^i(U) = H^{i-n}(\tilde{X}, \Omega_{\tilde{X}}^n(\log \tilde{D})).$$

This group vanishes for $i < n$, and also for $i > n$ because U is affine. For $i = n$, it is the space of n -forms on U with at most a simple pole along $\tilde{D}$. Since f is an isomorphism above a subset of X of codimension at least 2, such a form α has at most a simple pole along D at the generic points of D , hence everywhere. It must therefore be zero, which proves 3.2.

3.3. To use 3.1, one needs vanishing theorems for the cohomology of the twisted sheaves Ω_X^p . In the case of compact Hermitian symmetric spaces, the results of Bott [3] and Kostant [12] reduce the required vanishing to a combinatorial problem (cf. D. M. Snow [14,15]).

The irreducible compact Hermitian symmetric spaces are necessarily of the form $X := G/P$ with G simple and P a proper maximal parabolic. Their Picard group is $\mathbf{Z}$; we denote by $\mathcal{O}(1)$ its ample generator. Let $n = \dim X$. The invertible sheaf Ω_X^n is antiample, of the form $\mathcal{O}(-k)$; the integer k is computed in A. Borel and F. Hirzebruch [2, §16]. Their table shows that k is the dual Coxeter number of G , and in particular depends only on the type of G :

$$\begin{array}{cccccc} A_l & B_l & C_l & D_l & E_6 & E_7 \\ l+1 & 2l-1 & l+1 & 2l-2 & 12 & 18. \end{array}$$

3.4. The case where X is the Grassmannian $G(r, s)$ of r -dimensional subspaces in a vector space of dimension $r+s$, with $r, s > 0$, is the case where $G = \mathrm{SL}(r+s)$, hence of type A_l with $l = r+s-1$, and where P is the parabolic stabilizing an r -dimensional subspace. Its dimension is $n = rs$.

For every subset $R \subset [1, r+s] \subset \mathbf{Z}$ of cardinality r , let $\bar{R}$ denote its complement and define

$$l(R) := \#\{(i, j) \mid i \in R, j \in \bar{R}, j < i\}.$$

The symmetric group on $[1, r+s]$ is the Weyl group of type A_l , and $l(R)$ is the length of the permutation σ of $[1, r+s]$ that is increasing on $[1, r]$ and on $[r+1, r+s]$, with $\sigma([1, r]) = R$ and $\sigma([r+1, r+s]) = \bar{R}$.

Proceeding as in D. M. Snow [14], one deduces from Bott [3] and Kostant [12] that the following condition is necessary and sufficient for the vanishing of $H^*(X, \Omega^p(-a))$.

3.4.1. For every subset R of cardinality r in $[1, r+s]$, with complement $\bar{R}$ and with $l(R) = p$, one has

$$(R+a) \cap \bar{R} \neq \emptyset.$$

Proposition 12 (3.5). *With the above notation:*

- (i) $H^*(G(r, s), \Omega^p(-a)) = 0$ for $0 < a < l - p$.
- (ii) *The same conclusion holds for $0 < a \leq l - p$, except in the following cases: $H^*(G(3, 3), \Omega^3(-2))$, $H^*(G(2, r), \Omega^{r/2}(1 + r/2))$ for r even, and similarly for $G(r, 2) \cong G(2, r)$.*

Proof. We use criterion 3.4.1. Let $R, \bar{R}$ be as in 3.4 with $l(R) = p$. Put $x = \inf(\bar{R}) - 1$ and $y = \sup(R)$. One has $[1, x] \subset R$, $x+1 \in \bar{R}$, hence $(R+a) \cap \bar{R} \neq \emptyset$ if $0 < a \leq x$; likewise $[y+1, r+s] \subset \bar{R}$, $y \in R$, hence $(R+a) \cap \bar{R} \neq \emptyset$ if $0 < a \leq r+s-y$.

Assume therefore that $a > x$ and $a > r+s-y$. Then

$$\inf((R+a) \cup \bar{R}) = \inf \bar{R} = x+1,$$

$$\sup((R+a) \cup \bar{R}) = \sup(R+a) = y+a,$$

and $(R+a) \cup \bar{R} \subset [x+1, y+a]$ has at most $y-x+a$ elements.

The pairs (i, j) with $j = x+1$, $i \in R - [1, x]$, together with the pairs (i, j) with $i = y$, $j \in \bar{R} - [y+1, r+s]$, satisfy $i \in R$, $j \in \bar{R}$, and $j < i$. The pair (y, x) is counted twice. Hence

$$l(R) \geq (r-x) + (s-(r+s-y)) - 1 = y-x-1.$$

Equality holds if and only if R has the form

$$(3.5.1) \quad R = [1, x] \cup [x+2, r] \cup \{y\}, \quad R \neq [1, r].$$

If condition (i) holds, then

$$0 < a < (r+s-1) - (y-x-1) = (r+s) - (y-x)$$

and therefore

$$\#((R+a) \cup \bar{R}) < (y-x) + [(r+s) - (y-x)] = r+s,$$

so $(R + a) \cap \bar{R} \neq \emptyset$, as required.

To prove (ii), if one of the inequalities $a \leq l - p$ or $l(R) \geq y - x - 1$ is strict, the preceding argument still applies. Thus we may assume, and do assume, that $a = (r + s) - (y - x)$ and $l(R) = y - x - 1$. In particular, R has the form (3.5.1). If $x = 0$, then $a \leq r + s - y$. If $y = r + s$, then $a \leq x$. These cases have already been treated, so we may assume that $x \neq 0$ and $y \neq r + s$. Then $1 \leq x \leq r - 1$ and $r + 1 \leq y \leq r + s - 1$.

Assume $(R + a) \cap \bar{R} = \emptyset$ and show that this leads precisely to the exceptional cases in (ii). The translation $z \mapsto z + a$ sends y to $(r + s) + x$, and if $i \in [1, r + s]$, then either $i + a \in [1, r + s]$ or else $i + a - (r + s) \in [1, x] \subset R$. Let qR and $q\bar{R}$ denote the images of R and $\bar{R}$ in $\mathbf{Z}/(r + s)$. Since $(R + a) \cap \bar{R} = \emptyset$, one has $qR + a \subset q\bar{R}$, hence $qR + a = q\bar{R}$ and $q\bar{R} + a = qR$.

Call the connected components of R (resp. of qR) the largest intervals (resp. images of intervals) contained in R (resp. in qR). The connected components of qR are the images of those of R . By (3.5.1), R has two or three connected components. Translation by a permutes the connected components of qR , sending the one containing y to the next one, namely the one containing x . The complement $q\bar{R}$ has the same number of connected components, permuted in the same way.

If R has three connected components, then the component containing y is reduced to y and every component has one element. The same holds for the components of $\bar{R}$: x is a component by itself and every component of $\bar{R}$ has one element. Hence $r = s = 3$, $R = \{1, 3, 5\}$, and $a = 2$. This is the first exceptional case. One has $l(R) = 3$.

Assume now that R has two connected components. There are two cases: either $x = r - 1$ or $y = r + 1$. If $x = r - 1$, the component containing y is reduced to y and every component has one element. Thus $R = \{1, y\}$ and translation by a permutes 1 and y in $\mathbf{Z}/(r + s)$: one gets $r = 2$, s even,

$$R = \left\{1, 2 + \frac{s}{2}\right\}.$$

If $y = r + 1$, the component of x in $\bar{R}$ is reduced to x ; likewise one gets r even, $s = 2$, and

$$R = \left[1, \frac{r}{2}\right] \cup \left[\frac{r}{2} + 2, r + 1\right].$$

This is the second exceptional case. □

By 3.1 this proposition has the following corollary.

Corollary 13 (3.6). *Let $D \subset G(r, s)$ be the divisor defined by the vanishing of a section of $\mathcal{O}(d)$, and let U be the complementary open subset. Put $\mu = [(r + s - 1)/d]$. Except in the exceptional cases below, $H_c^*(U)$ has Hodge filtration $\geq \mu$. The exceptions are $G(3, 3)$ and $G(2, r) \cong G(r, 2)$ with $d = 1$, where the Hodge filtration is only $\geq \mu - 1$.*

Remark 14 (3.7). In the special case $X = \mathbf{P}^n$, Theorem 2 shows that a cohomology class in $H^*(U, \mathbf{C})$ can be represented by a holomorphic differential form on U with controlled pole order along D . Dimca uses this fact in [5, 6] to relate the Betti numbers and Alexander polynomial of a hypersurface with isolated singularities to the defect of associated linear systems.

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